

- 1 Let $\mathcal{M}$ be a σ -algebra and $(U_n)_{n \in \mathbb{N}} \subset \mathcal{M}$ be sequence contained in $\mathcal{M}$ with the property that all U_n are distinct. We want to show that $\mathcal{M}$ is uncountable. Consider the sequence $C_1 = U_1, C_2 = U_2 \setminus U_1, C_m = U_m \setminus \bigcup_{i=1}^{m-1} U_i$. Note that since the sets are distinct then that implies there exists for every $j \in \mathbb{N}$ a unique $x_j \in X$ such that $x_j \in U_j$. Thus $\emptyset \neq \{x_j\} \subset C_j$. We claim that $C_j \cap C_i = \emptyset$ for all $i, j \in \mathbb{N}, i \neq j$. Assume WLOG that $i < j$. Then C_j explicitly have any elements in U_i . Since $C_i \subseteq U_i$ then $C_i \cap C_j = \emptyset$. Now since each C_i is guaranteed to be unique, then we can construct the bijection between countable binary sequences and the countable union of elements in (C_i) . Note that this is achieved by assigning each binary digit, b_i , to whether C_i is included in the union. Note that by the distinctness of each element in (C_i) that each union is unique. Since the set of countable binary sequences is uncountable, then the set of countable union of elements from the constructed sequence is countable. Therefore since our constructed sequence is in $\mathcal{M}$ then $\mathcal{M}$ necessarily contains an uncountable number of elements. It can't have more than an uncountable number since only countable unions can be taken.

- 4 Let $(X, \mathcal{M})$ be a measurable space and $f : X \rightarrow \mathbb{R}$. We want to show that if $A \cup B = X, A, B \in \mathcal{M}$, then f is $\mathcal{M}, \mathcal{B}_{\mathbb{R}}$ measurable iff the restriction of f to A and B is measurable too.
 - $\Rightarrow$ Let f be $\mathcal{M}, \mathcal{B}_{\mathbb{R}}$ measurable. Note that the restriction to A or B is just a subset of $\mathcal{M}$ with the restriction that $f_A^{-1}(S) \subseteq A$. If values that are outside of the range of f_A are asked of it then this still doesn't affect what f_A^{-1} returns. Thus for any $S \in \mathcal{B}_{\mathbb{R}}, f_A^{-1}((a, b)) = f^{-1}((a, b)) \cap A \in \mathcal{A}$, where $\mathcal{A}$ is the algebra over A with respect to $\mathcal{M}$. Note that the same applies to B .
 - $\Leftarrow$. For all $S \in \mathcal{B}_{\mathbb{R}}, f_A^{-1}(S) \in \mathcal{A}, f_B^{-1}(S) \in \mathcal{B}, \mathcal{A}, \mathcal{B}$ are the algebras over A, B contained in $\mathcal{M}$. Note that the inverse of the restriction is equivalent to taking the intersection of the function inverse and the chosen set. Therefore $f^{-1}(S) = f^{-1}(S) \cap X = f^{-1}(S) \cap (A \cup B) = f^{-1}(S) \cap A \cup f^{-1}(S) \cap B = f_A^{-1}(S) \cup f_B^{-1}(S)$, which both of those are in $\mathcal{M}$, thus $f^{-1}(S) \in \mathcal{M}$

- 5 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be monotone. Note that f is discontinuous in only countably many places. Let the continuous portions (on their disjoint open set restriction) be denoted by the sequence $\{f_n\}_{n \in \mathbb{N}}$, allowing us to write $f(x) = \bigcup_{i=1}^{\infty} f_i(x)$. Let $S \in \mathcal{B}_{\mathbb{R}}$. Consider $f^{-1}(S)$. Note that by our definition $f^{-1}(x) = \bigcup_{i=1}^{\infty} f_i^{-1}(x)$. Since each f_i^{-1} is continuous on it's restriction, which is an open set in $\mathbb{R}$, then each $f_i^{-1}(S) \in \mathcal{B}_{\mathbb{R}}$, thus the countable union of sets in the borel algebra is in the borel algebra. Therefore $f^{-1}(S) \in \mathcal{B}_{\mathbb{R}}$

- 7 Let $(X, \mathcal{M}, \mu)$ be a measure space with $\mu(X) < \infty, (f_n)_{n \in \mathbb{N}}$ be given by $f_i : \mathcal{M} \rightarrow \mathbb{R}$ as a sequence of measurable functions, and suppose there exists $E \in \mathcal{M}, \mu(E) = 0$, where for all $x \in E^c, \sup_{n \in \mathbb{N}} |f_n(x)| < \infty$. Consider the sets $S_n = \{x : \sup_{n \in \mathbb{N}} |f_n(x)| > n\}$ where $n \in \mathbb{N}$. Note that $S_{i+1} \subseteq S_i$ since it is possible that there exists elements $x \in S_i$ such that $\sup_{n \in \mathbb{N}} |f_n(x)| < i + 1$. Therefore $\lim \mu(S_n) = \mu(\lim S_n)$. Note that if $x \in X, \sup_{n \in \mathbb{N}} |f_n(x)| = \infty$ then necessarily, $x \in E$. Additionally this implies that $x \in \bigcap_{i=1}^{\infty} S_i$. Since $\bigcap_{n=1}^{\infty} S_n = \lim S_n$, then necessarily $\lim S_n \subseteq E$. Then $\mu(\lim S_n) \leq \mu(E) = 0$, thus $\lim \mu(S_n) = 0$, which implies for all $\epsilon > 0$ we can always find a n

sufficiently large such that $\mu(S_n) < \epsilon$. Note that if $\mu(X) < \infty$ is relaxed then we can take $X = \mathbb{N}$ and have $\mu(S) = |S|$. Note that the functions $f_n(x) = \begin{cases} x & x < n \\ n & x \geq n \end{cases}$ are measurable functions since any open set in $\mathbb{R}$ containing a natural number between 1 and n will just include that in the inverse image of f_n , and since all finite subsets are in the σ algebra of $\mathbb{N}$. Additionally, for each $y \in \mathbb{N}$, $\sup_{n \in \mathbb{N}} |f_n(y)| = y$. Therefore $E = \emptyset$. However for any $C > 0$, $\mu(\{x \in \mathbb{N} : \sup_{n \in \mathbb{N}} |f_n(x)| > C\}) = \infty$.

- 8 Let $(X, \mathcal{M}, \mu)$ be a measure space with $\mu(X) < \infty$. Let $\{E_n\}_{n \in \mathbb{N}} \subset \mathcal{M}$, and let E be the set of all points such that $x \in E_n$ for an infinite number of n , and let $\mu(E) = 0$. Let $F_1 = \bigcup_{n=1}^{\infty} E_n$, and $F_i = \bigcup_{n=i}^{\infty} E_n$. Note that $F_{i+1} \subseteq F_i$ and $\bigcap_{i=1}^{\infty} F_i = \bigcap_{i=1}^{\infty} E_i$. Since $x \in \bigcap_{i=1}^{\infty} E_i$ implies that $x \in E$ then $\bigcap_{i=1}^{\infty} E_i \subset E$. Therefore $\lim \mu(E_n) = \mu(\bigcap_{n=1}^{\infty} E_n) \leq \mu(E) = 0$, which implies that $\lim \mu(E_n) = 0$. If the condition that $\mu(X) < \infty$ is relaxed then taking $X = \mathbb{R}$ we have the sequence $E_n = (n, n+1)$. Note that $E = \emptyset$, thus $\mu(E) = 0$, however $\mu(E_n) = 1$.